Complete schematic - sheet 1

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

BT1 | 3S LiPo / connector

INVENTORY / VERIFY

11.1 V nominal / 3300 mAh

PACK+	PACK_RAW
PACK-	GND
BAL_CELL0	CELL0
BAL_CELL1	CELL1
BAL_CELL2	CELL2
BAL_CELL3	CELL3

TB_GND | Common ground distribution

ADDITIONAL / MISSING OR SELECT

Conductor / terminal rating TBD

BUS	GND

S2 | Latching E-stop / 3 isolated NC contacts

ADDITIONAL / MISSING OR SELECT

SELECT PART

COIL_COM	COIL_PG2
COIL_NC	COIL_STOP_CHAIN
SENSE_COM	V3V3
SENSE_NC	SENSE_STOP_OK
EN_COM	ENABLE_5V
EN_NC	EN_DRIVER_5V

Z_COIL | Coil suppression network

ADDITIONAL / MISSING OR SELECT

SELECT FOR K1

A1	COIL_HOLD
A2	GND

Z_L | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	LEFT_SUPPLY
-	GND

F0 | main pack fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_RAW
OUT	PACK_FUSED

P1 | 3S high-side protection assembly

ADDITIONAL / MISSING OR SELECT

SELECT PART

IN+	PACK_ISOLATED
OUT+	PACK_PROTECTED
GND	GND
CELL0	CELL0
CELL1	CELL1
CELL2	CELL2
CELL3	CELL3

S3 | Momentary motor-power ARM

ADDITIONAL / MISSING OR SELECT

NO; rating TBD

COM	COIL_STOP_CHAIN
NO	COIL_HOLD

OV1 | 5V rail 1 OVP/OCP and supervision assembly

ADDITIONAL / MISSING OR SELECT

SELECT / VERIFY

IN+	V5_MCU_RAW
OUT+	V5_MCU
GND	GND
PG_COM	COIL_FUSED
PG_NO	COIL_PG1

Z_R | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	RIGHT_SUPPLY
-	GND

S0 | Master load isolator

ADDITIONAL / MISSING OR SELECT

DC rating TBD

COM	PACK_FUSED
OUT	PACK_ISOLATED

TB_POS | Protected positive distribution

ADDITIONAL / MISSING OR SELECT

Conductor / terminal rating TBD

BUS	PACK_PROTECTED

K1 | DC motor contactor

ADDITIONAL / MISSING OR SELECT

CONTACT / COIL TBD

MAIN_COM	PACK_PROTECTED
MAIN_NO	MOTOR_SWITCHED
A1	COIL_HOLD
A2	GND
AUX_HOLD_COM	COIL_STOP_CHAIN
AUX_HOLD_NO	COIL_HOLD
AUX_SENSE_COM	SENSE_STOP_OK
AUX_SENSE_NO	ESTOP_OK

OV2 | 5V rail 2 OVP/OCP and supervision assembly

ADDITIONAL / MISSING OR SELECT

SELECT / VERIFY

IN+	V5_LOGIC_RAW
OUT+	V5_LOGIC
GND	GND
PG_COM	COIL_PG1
PG_NO	COIL_PG2

Z_A | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	AUX_SUPPLY
-	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 2

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

F1 | MCU buck input fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_PROTECTED
OUT	MCU_BUCK_IN

BUCK2 | LM2596 #2

INVENTORY / VERIFY

5.0 V target

IN+	LOGIC_BUCK_IN
IN-	GND
OUT+	V5_LOGIC_RAW
OUT-	GND

FA | auxiliary driver fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	MOTOR_SWITCHED
OUT	AUX_SUPPLY

BUCK1 | LM2596 #1

INVENTORY / VERIFY

5.0 V target

IN+	MCU_BUCK_IN
IN-	GND
OUT+	V5_MCU_RAW
OUT-	GND

FL | left driver fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	MOTOR_SWITCHED
OUT	LEFT_SUPPLY

F2 | logic buck input fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_PROTECTED
OUT	LOGIC_BUCK_IN

FK | contactor coil fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_PROTECTED
OUT	COIL_FUSED

FR | right driver fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	MOTOR_SWITCHED
OUT	RIGHT_SUPPLY

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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Complete schematic - sheet 3

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

ESP1 | ESP32-WROOM-32 DevKit

INVENTORY / VERIFY

module terminal symbol

5V_VIN	V5_MCU
3V3	V3V3
GND	GND
GPIO16	L_FWD_3V3
GPIO17	L_REV_3V3
GPIO18	R_FWD_3V3
GPIO19	R_REV_3V3
GPIO14	ENABLE_3V3
GPIO21	BRUSH_IN1
GPIO22	BRUSH_IN2
GPIO23	CONVEYOR_IN3
GPIO25	CONVEYOR_IN4
GPIO32	TF_3V3
GPIO34	EF_SAFE
GPIO33	TL_3V3
GPIO35	EL_SAFE
GPIO13	TR_3V3
GPIO36	ER_SAFE
GPIO27	ESTOP_OK
GPIO0	NC - INSULATE
GPIO2	NC - INSULATE
GPIO4	NC - INSULATE
GPIO5	NC - INSULATE
GPIO12	NC - INSULATE
GPIO15	NC - INSULATE
GPIO26	NC - INSULATE
GPIO39	NC - INSULATE
UART0_TX_GPIO1	NC - INSULATE
UART0_RX_GPIO3	NC - INSULATE
EN_RESET	NC - INSULATE

US_L | HC-SR04 left

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TL_5V
ECHO	EL_RAW
GND	GND

U1 | SN74AHCT244N 3.3-to-5 V buffer

ADDITIONAL / MISSING OR SELECT

SN74AHCT244N

1 / 1OE_N	GND
2 / 1A1	L_FWD_3V3
3 / 2Y4	TR_5V
4 / 1A2	L_REV_3V3
5 / 2Y3	TL_5V
6 / 1A3	R_FWD_3V3
7 / 2Y2	TF_5V
8 / 1A4	R_REV_3V3
9 / 2Y1	ENABLE_5V
10 / GND	GND
11 / 2A1	ENABLE_3V3
12 / 1Y4	R_REV_5V
13 / 2A2	TF_3V3
14 / 1Y3	R_FWD_5V
15 / 2A3	TL_3V3
16 / 1Y2	L_REV_5V
17 / 2A4	TR_3V3
18 / 1Y1	L_FWD_5V
19 / 2OE_N	GND
20 / VCC	V5_LOGIC

DRV_A | ONE L298N: A brushes / B conveyor

INVENTORY / VERIFY

module terminal symbol

Vs_12V	AUX_SUPPLY
5V_LOGIC	V5_LOGIC
GND	GND
IN1	BRUSH_IN1
IN2	BRUSH_IN2
IN3	CONVEYOR_IN3
IN4	CONVEYOR_IN4
ENA	EN_DRIVER_5V
ENB	EN_DRIVER_5V
OUT1	DRV_A_OUT1
OUT2	DRV_A_OUT2
OUT3	DRV_A_OUT3
OUT4	DRV_A_OUT4

DRV_L | BTS7960 / IBT-2 left

INVENTORY / VERIFY

module terminal symbol

B+	LEFT_SUPPLY
B-	GND
M+	DRV_L_MP
M-	DRV_L_MN
VCC	V5_LOGIC
GND	GND
RPWM	L_FWD_5V
LPWM	L_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

DRV_R | BTS7960 / IBT-2 right

INVENTORY / VERIFY

module terminal symbol

B+	RIGHT_SUPPLY
B-	GND
M+	DRV_R_MP
M-	DRV_R_MN
VCC	V5_LOGIC
GND	GND
RPWM	R_FWD_5V
LPWM	R_REV_5V
R_EN	EN_DRIVER_5V
L_EN	EN_DRIVER_5V
R_IS	NC - INSULATE
L_IS	NC - INSULATE

US_F | HC-SR04 front

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TF_5V
ECHO	EF_RAW
GND	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 4

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

US_R | HC-SR04 right

INVENTORY / VERIFY

module terminal symbol

VCC	V5_LOGIC
TRIG	TR_5V
ECHO	ER_RAW
GND	GND

M5 | brush 1 motor

INVENTORY / VERIFY

12 V

A	M	M5_A
B		DRV_A_OUT2

CAM1 | ESP32-CAM - later phase

INVENTORY / VERIFY

reserved terminal symbol

5V	NC - INSULATE
GND	NC - INSULATE
U0R	NC - INSULATE
U0T	NC - INSULATE

FM3 | motor 3 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_R_MP
OUT	M3_A

FM6 | motor 6 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT1
OUT	M6_A

R24 | ECHO lower resistor

ADDITIONAL / MISSING OR SELECT

3.3k 1%, >=1/8 W

1	EF_SAFE
2	GND

M1 | left front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M1_A
B		DRV_L_MN

M2 | left rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M2_A
B		DRV_L_MN

M3 | right front wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M3_A
B		DRV_R_MN

M4 | right rear wheel motor

INVENTORY / VERIFY

12 V / 30 RPM

A	M	M4_A
B		DRV_R_MN

M6 | brush 2 motor

INVENTORY / VERIFY

12 V

A	M	M6_A
B		DRV_A_OUT2

M7 | conveyor motor

INVENTORY / VERIFY

Voltage TBD

A	M	M7_A
B		DRV_A_OUT4

FM1 | motor 1 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_L_MP
OUT	M1_A

FM2 | motor 2 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_L_MP
OUT	M2_A

FM4 | motor 4 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_R_MP
OUT	M4_A

FM5 | motor 5 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT1
OUT	M5_A

FM7 | motor 7 fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	DRV_A_OUT3
OUT	M7_A

R23 | ECHO upper resistor

ADDITIONAL / MISSING OR SELECT

2.2k 1%, >=1/8 W

1	EF_RAW
2	EF_SAFE

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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Complete schematic - sheet 5

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E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

R25 | ECHO upper resistor
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — EL_RAW
2 — EL_SAFE

R26 | ECHO lower resistor
ADDITIONAL / MISSING OR SELECT
3.3k 1%, >=1/8 W
1 — EL_SAFE
2 — GND

R27 | ECHO upper resistor
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — ER_RAW
2 — ER_SAFE

R28 | ECHO lower resistor
ADDITIONAL / MISSING OR SELECT
3.3k 1%, >=1/8 W
1 — ER_SAFE
2 — GND

R1 | L_FWD input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_FWD_3V3
2 — GND

R2 | L_REV input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_REV_3V3
2 — GND

R3 | R_FWD input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_FWD_3V3
2 — GND

R4 | R_REV input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_REV_3V3
2 — GND

R5 | ENABLE input pull-down
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — ENABLE_3V3
2 — GND

R6 | TF input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TF_3V3
2 — GND

R7 | TL input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TL_3V3
2 — GND

R8 | TR input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TR_3V3
2 — GND

R9 | BRUSH_IN1 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — BRUSH_IN1
2 — GND

R10 | BRUSH_IN2 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — BRUSH_IN2
2 — GND

R11 | CONVEYOR_IN3 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — CONVEYOR_IN3
2 — GND

R12 | CONVEYOR_IN4 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — CONVEYOR_IN4
2 — GND

R13 | DRV_L.R_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R14 | DRV_L.L_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.

Complete schematic - sheet 6

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E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

R15 | DRV_R.R_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R16 | DRV_R.L_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R17 | DRV_A.ENA local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R18 | DRV_A.ENB local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R19 | L_FWD receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_FWD_5V
2 — GND

R20 | L_REV receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_REV_5V
2 — GND

R21 | R_FWD receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_FWD_5V
2 — GND

R22 | R_REV receiver pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_REV_5V
2 — GND

R29 | E-stop feedback pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — ESTOP_OK
2 — GND

C_U1 | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_F | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_L | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_R | local ceramic bypass
ADDITIONAL / MISSING OR SELECT
100nF, >=10V
1 — V5_LOGIC
2 — GND

C_BL | local motor-bus reservoir
ADDITIONAL / MISSING OR SELECT
C / ESR / voltage TBD
+ — LEFT_SUPPLY
- — GND

C_BR | local motor-bus reservoir
ADDITIONAL / MISSING OR SELECT
C / ESR / voltage TBD
+ — RIGHT_SUPPLY
- — GND

C_BA | local motor-bus reservoir
ADDITIONAL / MISSING OR SELECT
C / ESR / voltage TBD
+ — AUX_SUPPLY
- — GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

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